SC7 Bayes Methods MT26

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Lecture 10 : Model Averaging

Model averaging

We can allow for uncertainty in which model is the right model when we estimate a function $h(\theta)$ of the parameters θ.

Suppose $\theta \in \Omega_m$ when the true model is $M = m$, with $m \in \mathcal{M}$. The extended parameter space including model index is

$$\Omega^* = \bigcup_{m \in \mathcal{M}} \{m\} \times \Omega_m.$$

Instead of selecting a model and estimating $h(\theta)$ in that model, we integrate over the model uncertainty and work with

$$\pi(\theta|y) = \sum_{m \in \mathcal{M}} \pi(\theta, m|y) \qquad \text{(A)}$$

Here $\pi(\theta, m|y)$ is a distribution on Ω^* and $\pi(\theta|y)$ on the space

$$\Omega = \bigcup_{m \in \mathcal{M}} \Omega_m.$$

We use density notation. Implicitly the model spaces have the same measure and only differ by densities, $\pi(\theta|m)$ and $p(y|\theta, m)$.

In terms of our old model distributions

$$\pi(\theta|y) = \sum_m \pi(\theta|m,y)\pi(m|y) \qquad \text{(B), or alternatively,}$$

$$= \sum_m \frac{p(y|\theta,m)\pi(\theta|m)\pi(m)}{p(y)} \qquad \text{(C), directly from (A).}$$

Here $\pi(\theta|m,y), p(y|\theta,m)$ and $\pi(\theta|m)$ are the old single-model distributions. In the model posterior in (B),

$$\pi(m|y) = \frac{p(y|m)\pi(m)}{p(y)},$$

the normalising constant is $p(y) = \sum_m p(y|m)\pi(m)$ with

$$p(y|m) = \int_{\Omega_m} p(y|\theta,m)\pi(\theta|m)d\theta, \quad m \in \mathcal{M},$$

the old within-model marginal likelihoods. The joint distribution

$$\pi(\theta,m|y) \propto p(y|\theta,m)\pi(\theta|m)\pi(m), \qquad (\theta,m) \in \Omega^*$$

has normalising constant $p(y)$ above. This form is tractable.

When different model-specific parameters are needed below,

$$\pi(d\theta|y) = \sum_{m\in\mathcal{M}} \pi(d\theta|m,y)\pi(m|y)$$

and $\int_{\Omega_m} \pi(d\theta|m,y) = 1$.

This means that for measureable $A \subseteq \Omega$

$$\int_A \pi(d\theta|y) \equiv \sum_{m\in\mathcal{M}} \pi(m|y) \int_{A\cap\Omega_m} \pi(d\theta|m,y)$$

For example, if $\Omega_m = \mathbb{R}^{p_m}$ so $\theta \in \Omega_m$ has dimension p_m, then $d\theta$ is the element of volume in $\mathbb{R}^{p_m}$ in the distribution $\pi(d\theta|m,y)$.

Suppose $h_m : \Omega_m \to \mathcal{H}$ defines a quantity $H = h_m(\theta)$ taking values in the same space $\mathcal{H}$ for every model. Then

$$\Pr(H \in A \mid y) = \sum_m \pi(m \mid y) \int_{\Omega_m} \mathbb{I}_{\{h_m(\theta)\in A\}} \pi(\theta \mid m, y)\, d\theta.$$

Example (averaging over link functions)

Challenger O-ring data, with x_i scaled temperature, $y_i \in \{0, 1\}$,

$$y_i \sim \text{Bernoulli}(\mu(\beta_1 + \beta_2 x_i)) \; i = 1, ..., n,$$

$\mu = \mu_m(\eta),\ m \in \mathcal{M}$ with $\mathcal{M} = \{1, 2\}$, μ_1 logit, μ_2 probit link.

$$\pi(\beta, m|y) \propto p(y|\beta, m)\pi(\beta|m)\pi(m), \qquad (\beta, m) \in \Omega^*$$

The likelihoods are different in the two models.

$$p(y|\theta, m) = \begin{cases} \text{logit lkd} & \text{if } m = 1, \\ \text{probit lkd} & \text{if } m = 2. \end{cases}$$

$\pi(\beta|m) = N(\beta; 0, 9I_2)$ for $m = 1, 2$ and $\pi(m) = 1/2$ (say).

Using model averaging over link functions the model parameter spaces are equal, $\Omega_1, \Omega_2 = \mathbb{R}^2 = \Omega$ say.

$$\Omega^* = (\{1\} \times \mathbb{R}^2) \cup (\{2\} \times \mathbb{R}^2) = \mathcal{M} \times \Omega.$$

Simple product space as parameter spaces don't depend on model.

We want to learn about β averaged over models so want

$$\pi(\beta|y) = \sum_m \pi(\beta, m|y).$$

Could do MCMC targeting $\pi(\beta, m|y)$ then sample $\beta \sim \pi(\cdot|y)$.

Here we can actually calculate the marginal directly using (B). The model average posterior is

$$\pi(\beta|y) = \pi_1\, \pi(\beta|m=1, y) + \pi_2\, \pi(\beta|m=2, y)$$

with $\pi_m = \pi(m|y)$ for $m = 1, 2$. Since $\pi(m) = 1/2$ we have

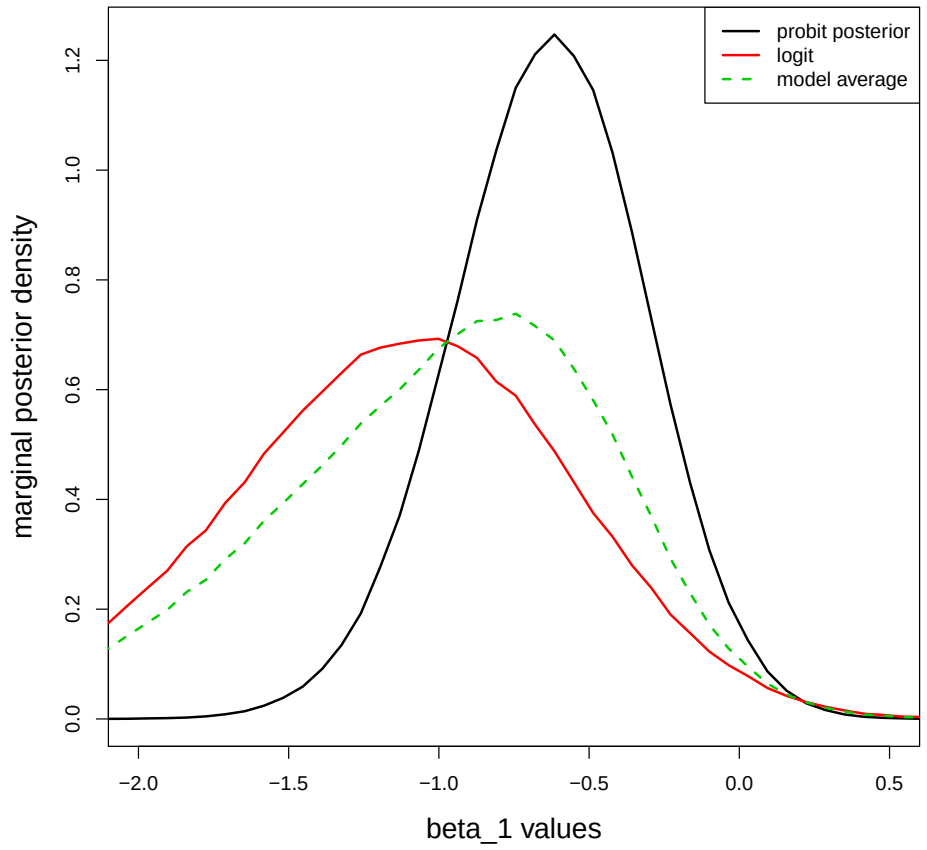
$$\begin{aligned} \pi_1 &= p(y|1)/(p(y|1) + p(y|2)) \\ &= B_{1,2}/(B_{1,2} + 1), \end{aligned}$$

in terms of the Bayes Factor $B_{1,2} \simeq 2.75$ we estimated before.

$$\pi(\beta|y) = \pi_1\,\pi(\beta|m=1,y) + (1-\pi_1)\,\pi(\beta|m=2,y)$$

Take MCMC samples from model posteriors $\pi(\beta|m,y),\ m=1,2$ and then resample the samples with probs $\pi_m = \pi(m|y)$.

Posterior densities for $\beta_1|y$ below The model-averaged posterior is weighted towards logit as it has a higher marginal likelihood.



Perspective from Decision Theory If our loss function for estimating $h(\theta)$ is the square error, the EPL is

$$\rho(\pi, \hat{h}|y) = \sum_{m \in \mathcal{M}} \int_{\Omega_m} (h(\theta) - \hat{h})^2 \pi(\theta, m|y) d\theta$$

and this is minimised by the model-averaged posterior mean

$$\begin{aligned} \hat{h}(y) &= E_{\theta, m|y}(h(\theta)) \\ &= \sum_{m \in \mathcal{M}} \int_{\Omega_m} h(\theta) \pi(\theta, m|y) d\theta, \end{aligned}$$

This is coherent inference minimising the Bayes risk.

Selecting a model $m^* = \arg\max_m p(y|m)$ first and using

$$E_{\theta|y,m^*}(h(\theta)) = \int_{\Omega_{m^*}} h(\theta) \pi(\theta|y, m^*) d\theta,$$

must have equal or higher risk. It doesn't allow for uncertainty in the chosen model m^*.

Model averaging using spike and slab priors

[see eg Hoff 9.3.1 - I use simulation and different example]

Consider model averaging in regression, with n observations and p covariate vectors $X = [X_1, \ldots, X_p]$,

$$Y \sim N(X\theta_z, \sigma^2 I_n), \quad \theta_z = (z_1\theta_1, ..., z_p\theta_p)$$

with $z = (z_1, ..., z_p)$ a vector of binary indicator variables, so

$$Y = \sum_{j=1}^{p} z_j X_j \theta_j + \epsilon, \quad \text{with } \epsilon \sim N(0, \sigma^2 I_n).$$

Here $z_j \in \{0, 1\}$ toggles the effect of the j'th covariate $X_j\theta_j$. For eg $z = (1, 1, 0, \ldots, 0)$ gives linear regression of Y on X_2.

There are 2^p models. Our model index $z \in \mathcal{M}$ takes values in

$$\mathcal{M} = \{0, 1\}^p.$$

The joint posterior is

$$\pi(\theta, \sigma, z|y) \propto p(y|\theta, \sigma, z)\pi(\theta, \sigma)\pi(z),$$

with $p(y|\theta, \sigma, z) = N(y; X\theta_z, \sigma^2 I_n)$ and $\pi(\theta, \sigma)$ and $\pi(z)$ respectively parameter and model priors.

The parameter space for $(\theta, \sigma) \in \Omega_z$ in model z is $\Omega_z = \Re^p \times \Re^+$ so it doesnt depend on the model z. The distribution $\pi(\theta, \sigma, z|y)$ is defined on the space

$$\Omega^* = \Re^p \times \Re^+ \times \mathcal{M}.$$

Remark 1: we can sample this posterior using our existing MCMC toolbox as all models have the same parameter space, $\mathbb{R}^p \times \mathbb{R}^+$.

If $z_i = 0$ then the posterior distribution for θ_i is just pure prior (θ_i doesn't appear in the likelihood when $z_i = 0$).

Remark 2: This θ-prior is called a "spike and slab" prior. Suppose

$$\pi(\theta, \sigma, z) = \pi_\sigma(\sigma) \prod_i \pi(\theta_i)\pi_z(z_i).$$

Let $w = \Pr(z_i = 0)$ in the z-prior. The "real" regression parameter $\tilde{\theta}_i = \theta_i z_i$ has prior CDF $\Pr(\tilde{\theta}_i \leq c)$ equal to

$$\Pr(\tilde{\theta}_i \leq c | z_i = 0) \Pr(z_i = 0) + \Pr(\tilde{\theta}_i \leq c | z_i = 1) \Pr(z_i = 1)$$

$$= w \, \mathbb{I}_{c \geq 0} + (1 - w) \int_{-\infty}^{c} \pi(\theta_i) d\theta_i$$

so the prior density is "spike" plus "slab",

$$\pi_{\tilde{\theta}}(\tilde{\theta}_i) = w\delta_0(\tilde{\theta}_i) + (1 - w)\pi(\tilde{\theta}_i),$$

in terms of the delta-function notation of lecture 3.

Model averaging for the **swiss** dataset in the R **MASS** package

I dropped two outliers and centred and scaled the response y. The five covariates are logits of percentages.

```
> sw.lm<-lm(Fertility~Infant.Mortality+Examination
+            +Education+Catholic+Agriculture, data=sw)
> summary(sw.lm)
...
Coefficients:
                 Estimate Std. Err ... Pr(>|t|)
(Intercept)       0.99    0.99     ... 0.324
Infant.Mortality  2.16    0.66     ... 0.002 **
Examination      -0.54    0.28     ... 0.061 .
Education        -0.48    0.20     ... 0.024 *
Catholic          0.08    0.05     ... 0.095 .
Agriculture      -0.23    0.14     ... 0.116
...
```

Priors: In this example we will take the priors as given.

The covariates are $O(1)$ so take $\theta_i \sim t(3)$ and $\pi(\sigma) \propto 1/\sigma$ in the parameter priors. The prior over models is

$$\pi_z(z) = \prod_{i=1}^{p} \xi^{z_i}(1-\xi)^{1-z_i},$$

with say $E(\sum_i z_i) = p\xi = c$ and c small, I use $c = 3$ for $p = 6$.

Posterior: Joint distribution of model index and parameters is

$$\pi(\theta, z, \sigma|y) \propto N(y; X\theta_z, \sigma^2 I_n) \times \prod_{i=1}^{p} t(\theta_i; df = 3) \times \sigma^{-1} \times \xi^{|z|}(1-\xi)^{p-|z|}$$

with $|z| = \sum_i z_i$.

MCMC targeting $\pi(\theta, z, \sigma|y)$*:* random-walk MH-MCMC. Suppose $X_t = (\theta, z, \sigma)$ is the Markov chain state at step t. Cycle through a θ-update, a z-update and a σ-update sequentially.

θ-update: choose $i \sim U\{1, 2, ..., p\}$ and simulate a proposal

$$\theta_i' \sim U(\theta_i - a, \theta_i + a).$$

The new regression parameters are

$$\theta_z' = (z_1\theta_1, ..., z_{i-1}\theta_{i-1}, z_i\theta_i', z_{i+1}\theta_{i+1}, , ..., z_p\theta_p).$$

Acceptance probability is

$$\alpha_{\theta_i}(\theta'|\theta) = \min\left\{1, \frac{N(y; X\theta_z', \sigma^2 I_n)t(\theta_i'; df)}{N(y; X\theta_z, \sigma^2 I_n)t(\theta_i; df)}\right\}.$$

Everything else cancels as other parameters are held fixed.

z-update: choose $i \sim U\{1, 2, ..., p\}$ and set $z_i' = 1 - z_i$ (all else unchanged) giving

$$\theta_{z'} = (z_1\theta_1, ..., z_{i-1}\theta_{i-1}, z_i'\theta_i, z_{i+1}\theta_{i+1}, ..., z_p\theta_p)$$

and

$$\alpha_{z_i}(\theta_{z'}|\theta_z) = \min\left\{1, \frac{N(y; X\theta_{z'}, \sigma^2 I_n)\xi^{z_i'}(1-\xi)^{1-z_i'}}{N(y; X\theta_z, \sigma^2 I_n)\xi^{z_i}(1-\xi)^{1-z_i}}\right\}.$$

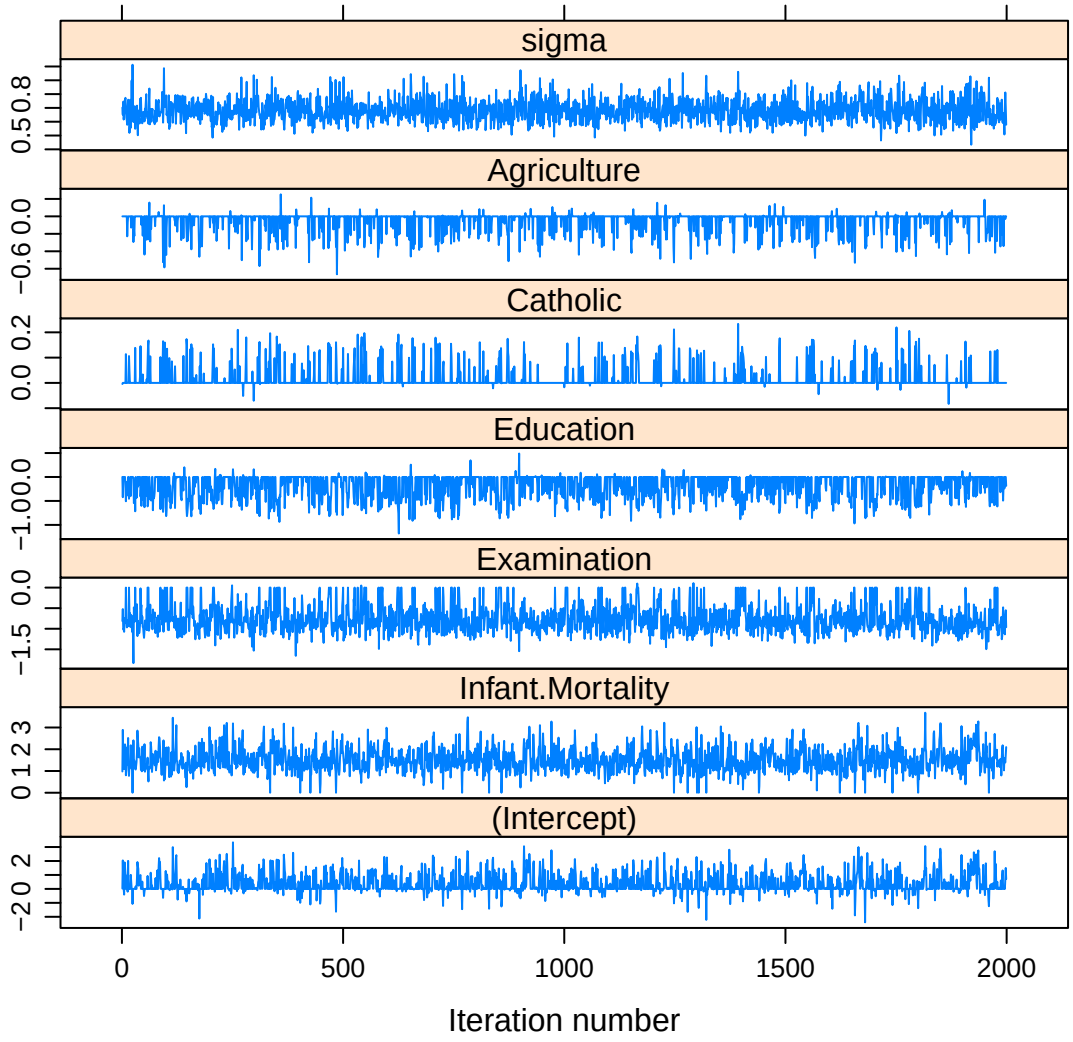
There is also a σ update. In the code online I use a "random walk on a log scale" (comes up in a later lecture).

The R-code is available on the course website and github.

We ran the code and generated $T = 10^7$ samples

$$(\theta^{(t)}, z^{(t)}, \sigma^{(t)}) \sim \pi(\theta, z, \sigma|y) \quad t = 1, 2, ..., T.$$

θ_z plots sub-sampled at 5000 steps. $ESS > 1000$ all parameters.



The posterior probabilities $\pi(z|y)$ can be estimated via

$$\hat{\pi}_z = T^{-1} \sum_t \mathbb{I}_{z^{(t)}=z}$$

Results as percentages: the 95% HPD set for models is

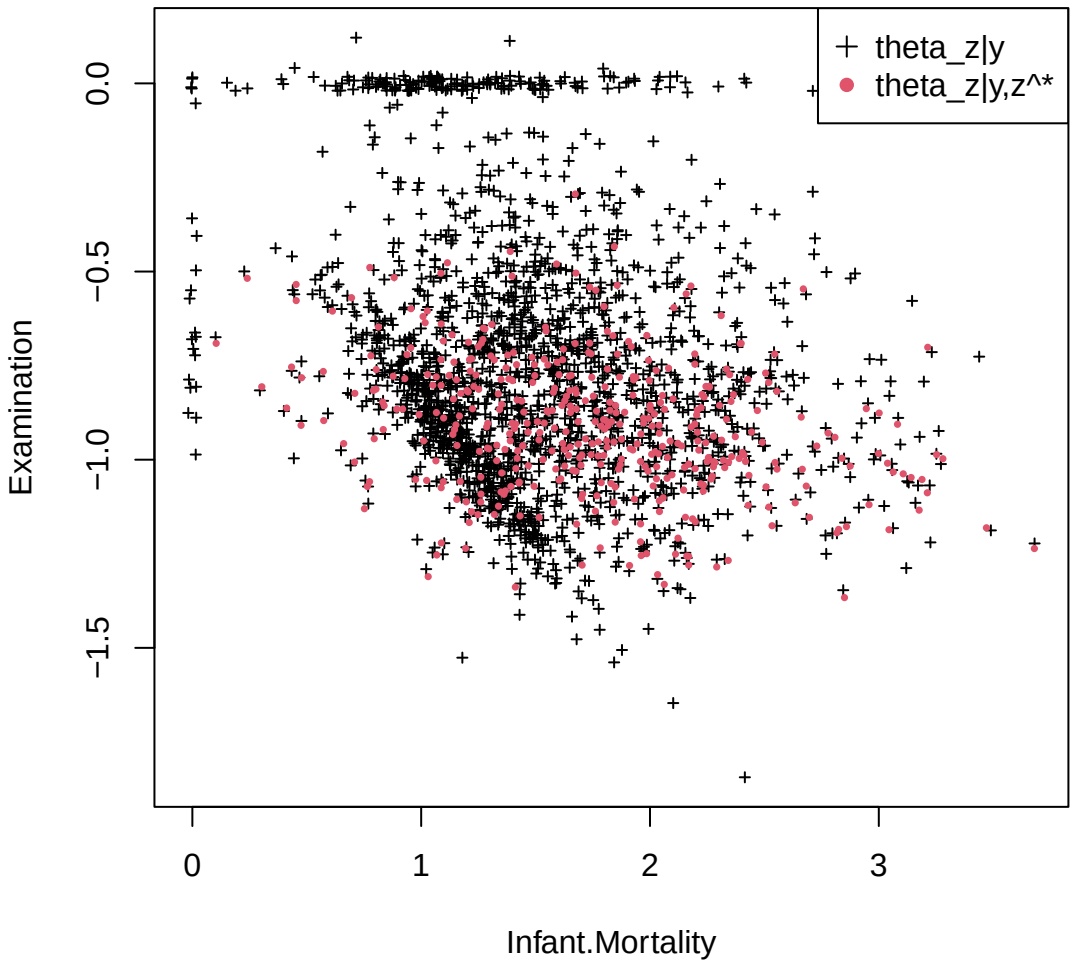
111000	011000	111100	011100	011101	111101	011001	111001
21.8	17.3	13.1	12.9	5.4	4.3	3.8	3.4
111110	011111	110110	010110	010111	011110	111111	111010
1.9	1.8	1.8	1.7	1.6	1.6	1.5	1.4

The MAP model is $z^* = (1, 1, 1, 0, 0, 0)$ with

intercept, **Infant.Mortality** and **Examination**.

The marginal probabilities are

Int	InfM	Exam	Edu	Cath	Agric
0.52	0.99	0.92	0.50	0.16	0.24



Red points: posterior conditioned on the MAP model

Black + signs: posterior allowing for uncertainty in the model.

The model averaged posterior gives better quantification of the uncertainty for downstream inference.